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SECP3213-02 - BUSINESS INTELLIGENCE

PHASE 3 : Visualization & Dashboard

TITLE : Retail Sales Performance and Customer Behaviour Analysis

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1. Introduction

This phase presents the completed Business Intelligence dashboards built in Microsoft Power BI using the star schema produced in Phase 2. Two interactive dashboards were developed, Dashboard 1 targets objective 1 by analysing sales revenue, average order value and product-category performance. Dashboard 2 targets objectives 2 and 3 by segmenting customers using RFM analysis and investigating the drivers of low customer review scores. A date-range slicer and a category dropdown slicer are synced across both dashboards to allow cross-page filtering. The dashboards are supported by five DAX measures that correctly account for the order-item grain of the fact table.

2. Dashboard 1: Sales & Profit Performance

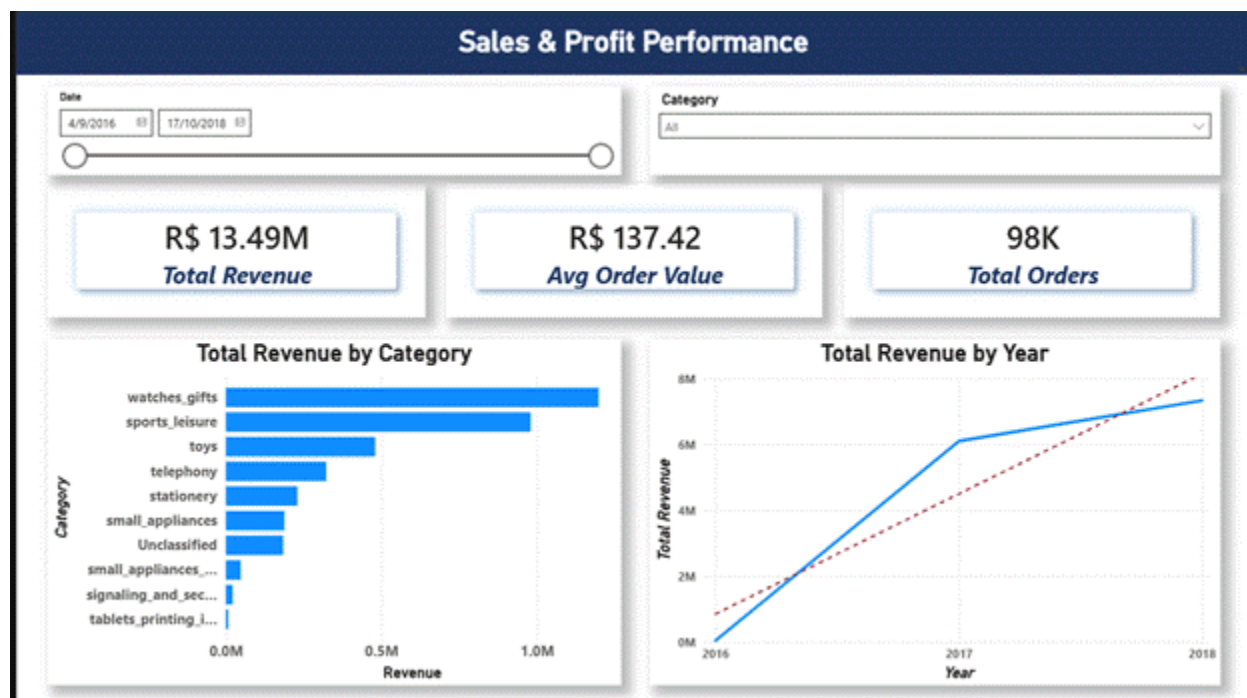


Figure 2.1 Dashboard 1: Sales & Profit Performance

2.1 Visualizations, Chart Justification and Best Practices

This dashboard shows how well the business is doing by putting three big numbers right at the top. At a quick glance, total sales reached R\$13.49M, the average order was R\$137.42, and customers placed about 98K orders. These simple cards were used so anyone can see the main results instantly without getting confused by too much data.

To see what brings in the most money, the dashboard uses a horizontal bar chart that lists products from highest sales to lowest. A horizontal chart was chosen because the product names are long, and this keeps them easy to read. By showing only the top 10 types of products, it keeps the chart clean. The numbers show that watches_gifts around R\$1.1M and sports_leisure around R\$0.8M are the best sold items, making the items a huge part of the total money.

To show how the business is growing, a line chart tracks sales from 2016 to 2018. A line chart is perfect here because the upward tilt of the line shows growth instantly. Sales went from almost nothing in 2016 to R\$6M in 2017 and hit R\$7.5M in 2018. The chart also adds a dashed red trend line to show normal, steady growth. Because the real sales line climbed above this red line in 2018, it proves that sales are actually speeding up, not just growing at a slow and steady pace.

Finally, there are easy-to-use filters at the top of the page for dates and product types. The dashboard provides selection for a specific date range or pick a certain product, every single number and chart on the page updates at the exact same time. This makes it incredibly easy to zoom in on a specific product or time frame without needing to click around to other pages.

2.2 Key Insights

Insight	Business implication
The business made R\$13.49M from 98K orders, averaging R\$137.42 per order.	Sales volume is good, but the business needs to find ways to get customers to buy more items per order.

Just two categories (watches/gifts and sports/leisure) bring in 14% of all money, out of 70+ total categories.	These two areas are super important. Running out of stock or having quality issues here will badly affect total sales.
Sales in 2018 hit R\$7.5M, beating the normal trend line.	Growth is speeding up. If this keeps up, sales could pass R\$10M next year.
"Unclassified" (unlabeled) products are sitting in the top 10 list.	There is a messy data problem. Labeling these items will show what is actually selling and help marketing.

3. Dashboard 2: Customer Behaviour & Reviews

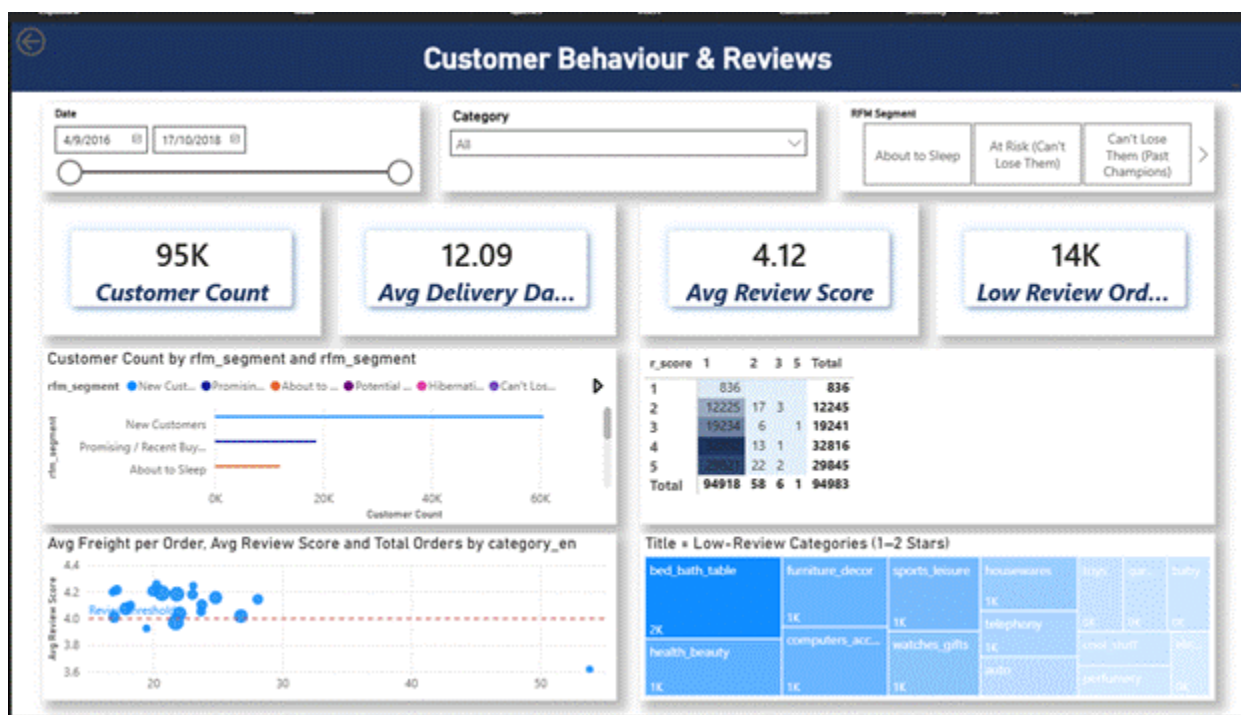


Figure 3.1 Dashboard 2: Customer Behaviour & Reviews

3.1 Visualizations, Chart Justification, and Best Practices

The dashboard highlights four key numbers to give you a quick overview of the business. There are 95,000 unique customers, an average delivery time of about 12 days and an average review score of 4.12 out of 5. However, there are 14,000 low-review orders (1 or 2 stars). This bad review number is included to highlight the main concern that needs to be fixed immediately

Next, a horizontal bar chart groups customers based on their shopping habits. It shows that "New Customers" make up the biggest group by far, with around 60,000 people. "Promising/Recent Buyers" come in second at roughly 20,000, and "About to Sleep" (customers who have not shopped in a while) sits at about 10,000. The chart is color-coded and any group clicked is instantly filtered with the rest of the dashboard and see details for just those customers.

This matrix chart uses a color coded grid to show when people last shopped compared to how much they spent. Darker colors mean more customers. The grid reveals a huge group of over 32,000 customers who bought something very recently but spent very little money. This is a good indicator because it means the customers are still active. Those customers can be targeted to try and encourage them to spend more on their next purchase. Customer buying frequency is not included because 97% of customers only bought something once which is not very insightful.

The scatter plot chart causes floating bubbles to see if high shipping fees lead to bad reviews. Each bubble is a product category with bigger bubbles meaning more orders. Most products cluster together with shipping costs between R\$20 and R\$35 and decent reviews between 4.0 and 4.4 stars. However, a category with very high shipping costs (around R\$50) that also has the lowest rating on the platform (3.6 stars) shows to be an outlier. This shows that expensive shipping is affecting customer satisfaction in that specific category.

Finally, a tree map highlights which product types get the most 1-star and 2-star reviews. The bigger and darker the box, the more unsatisfied customers there are. It only shows the top 15 worst categories so the chart is more targeted. "Bed, bath and table" is the biggest problem area with about 2,000 bad reviews. It is followed by furniture, sports gear, health/beauty items, computer accessories and watches which have 1,000 bad reviews each.

3.2 Key Insights

Insight	Business implication
About 63% of customers (around 60,000 out of 95,000) are brand new and have never bought anything a second time.	The company is great at getting new customers but struggles to keep them. Setting up a rewards program or sending follow-up messages is the best way to get them to return.
There is a large group of 32,816 customers who shopped quite recently but spent very little money.	These buyers are still new, so offers such as special deals, upgrades or product bundles are encouraged to get them to spend more.
The overall average review score is good at 4.12 out of 5 stars, but 14% of all orders (around 14,000) received bad 1 or 2-star reviews.	The good average hides a large group of very unhappy customers. Specific product categories need to be looked at carefully to identify the reason for bad review for those categories.
The "bed, bath and table" category has the most bad reviews (2,000 orders) which is twice as many as any other category.	The product quality and delivery process for these items need to be checked immediately. Addressing the complaints in half would save 1,000 customers from a bad experience.
The products with the highest shipping costs also have the lowest customer review scores.	Expensive shipping leads to unhappy customers. Negotiate by offering cheaper shipping rates or offer free shipping over a certain price for improved customer experience

4. Business Recommendations

Based on the combined analysis across both dashboards, the following four actions are recommended in order of estimated business impact:

- **Focus marketing and stock on watches and gifts and sports and leisure.** These two areas bring in a huge share of the R\$13.49M total revenue. Running out of stock or having bad quality here would hurt the business deeply.
- **Bring back one-time buyers.** Since 63% of customers only buy something once and never return, creating a campaign for new or recent buyers is an easy way to grow sales without spending too much money.
- **Fix the bed_bath_table category.** This group has about 2,000 bad 1-star and 2-star reviews, which is twice as many as any other category. Checking its quality and shipping is the fastest way to raise overall customer satisfaction.
- **Lower high shipping costs.** When shipping costs too much compared to the item's value, customers leave bad reviews. Finding cheaper delivery options or offering shipping discounts will make customers happier.
- **Clean up the missing product data.** "Unclassified" products are sitting in the top 10 list, which means a lot of sales money is hidden. Labeling these items correctly will show exactly what is selling and help the business make smarter choices.